

Practice Questions on Pointers:

1. Basic Pointer Operations

Write a program that demonstrates the following:

- Declare a pointer to an integer.
- Assign a value to the integer via the pointer.
- Print the value using both the pointer and the regular variable.
- Perform pointer arithmetic to increment the pointer and access other values in an array.

Task: Implement the program and print the results for each operation.

2. Pointer to an Array

Create a program where:

- Declare an array of integers.
- Use a pointer to traverse the array and print each element.
- Modify the elements of the array using the pointer.
- Print the modified array.

Task: Implement the program and show how pointer arithmetic can be used to traverse and modify the array.

3. Dynamic Memory Allocation

Write a program that:

- Dynamically allocates memory for an array of integers using `new`.
- Accepts user input to populate the array.
- Calculates and prints the sum of the elements.
- Frees the dynamically allocated memory using `delete[]`.

Task: Implement the program and ensure proper memory management.

4. Function with Pointer as Argument

Create a program with a function that accepts a pointer as an argument. The function should:

- Accept an integer pointer and increment the value at the pointer by 1.
- Print the value of the integer before and after the increment.

Task: Implement the program and demonstrate passing pointers to functions.

5. Pointer to Pointer

Write a program that:

- Declares a pointer to an integer.
- Declares a pointer to the pointer.
- Assigns a value to the integer via the pointer and prints the value through both pointers.

Task: Implement the program to show how a pointer can point to another pointer and demonstrate pointer-to-pointer operations.

6. Function Returning Pointer

Create a program with a function that:

- Accepts two integers as arguments.
- Returns a pointer to the larger of the two integers.
- Prints the returned pointer value and the integer it points to.

Task: Implement the function and demonstrate returning a pointer from a function.

7. Pointer and Strings

Write a program that:

- Declares a string and a pointer to a character.
- Use the pointer to iterate over each character of the string and print them one by one.
- Change a character in the string using the pointer and print the modified string.

Task: Implement the program and demonstrate how pointers can be used with strings in C++.

8. Pointer Arithmetic

Write a program that:

- Declares an array of integers.
- Creates a pointer to the first element of the array.
- Use pointer arithmetic to print the array's elements by incrementing the pointer and accessing each element.

Task: Implement the program to show how pointer arithmetic can be used to access array elements.

9. NULL Pointer

Create a program that:

- Declares a pointer and initializes it to **NULL**.
- Check if the pointer is **NULL** and print an appropriate message.
- Allocate memory dynamically for an integer, assign it a value, and print the value.

Task: Implement the program and demonstrate the usage of **NULL** pointers and dynamic memory allocation.

10. Pointer to Structure

Write a program that:

- Define a structure **Student** with fields: **name**, **age**, and **grade**.
- Declare a pointer to a structure **Student**.
- Use the pointer to initialize the structure and access its members.
- Print the details of the structure using the pointer.

Task: Implement the program and show how pointers can be used with structures.

11. Passing Array to Function Using Pointers

Create a program where:

- A function accepts an array and its size as parameters.
- The function should compute and return the average of the array elements.
- Use a pointer to access the array elements in the function.

Task: Implement the program and ensure correct usage of pointers with arrays passed to functions.

12. Pointer to a Function

Write a program that:

- Define a function that adds two integers.
- Declare a pointer to the function and use the pointer to call the function.

Task: Implement the program and show how function pointers can be used.

15. Swap Using Pointer

Write a program that:

- Defines a function `swap(int *a, int *b)` that swaps the values of two integers using pointers.
- Call the function in the `main` function with two integer variables.

Task: Implement the program and demonstrate how pointers can be used to swap values in memory.
